

Multiple Choice

1. **Answer:** A

Options: A) Reference counting is well suited for reclaiming cyclic structures.; B) Reference counting incurs additional space overhead for each memory cell.; C) Reference counting is an alternative to mark-and-sweep garbage collection.; D) Reference counting need not keep track of which cells point to other cells.

Note: Correct option: A - Reference counting is well suited for reclaiming cyclic structures.

2. **Answer:** D

Options: A) $O(\log N)$; B) $O(N \log N)$; C) $O(\log N) + O(1)$; D) $O((\log N)^2)$

Note: Correct option: D - $O((\log N)^2)$

3. **Answer:** D

Options: A) choice(seq); B) randrange ([start,] stop [,step]); C) random(); D) seed([x])

Note: Correct option: D - seed([x])

4. **Answer:** A

Options: A) xyz; B) xy; C) xxzy; D) xxxxy

Note: Correct option: A - xyz

5. **Answer:** B

Options: A) 3; B) 4; C) 2; D) 8

Note: Correct option: B - 4

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'a[i] == max'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def text_match(text): | return ('Found a match!') | return ('Not matched!')`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def surfacearea_cube(l): | return surfacearea`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key